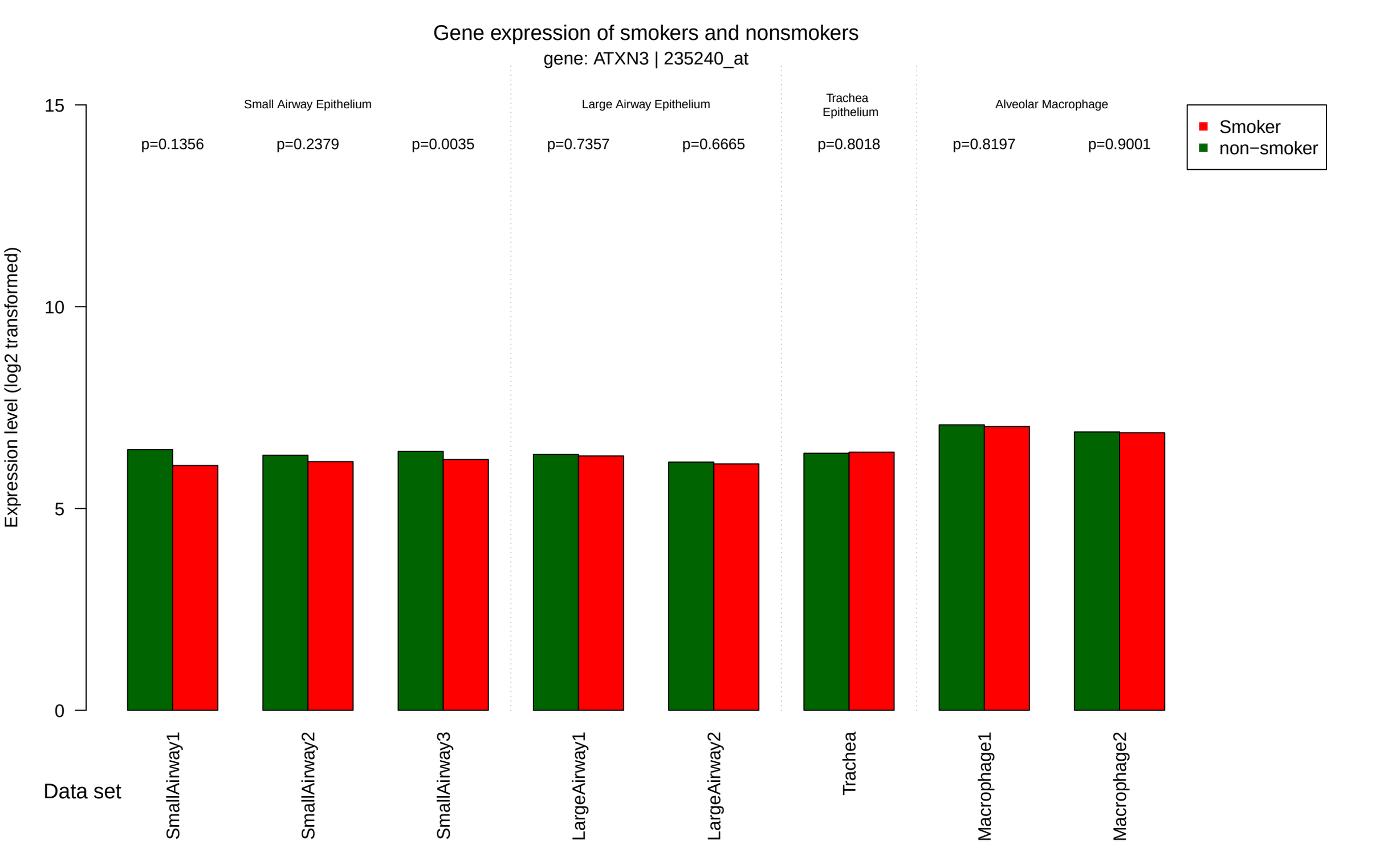
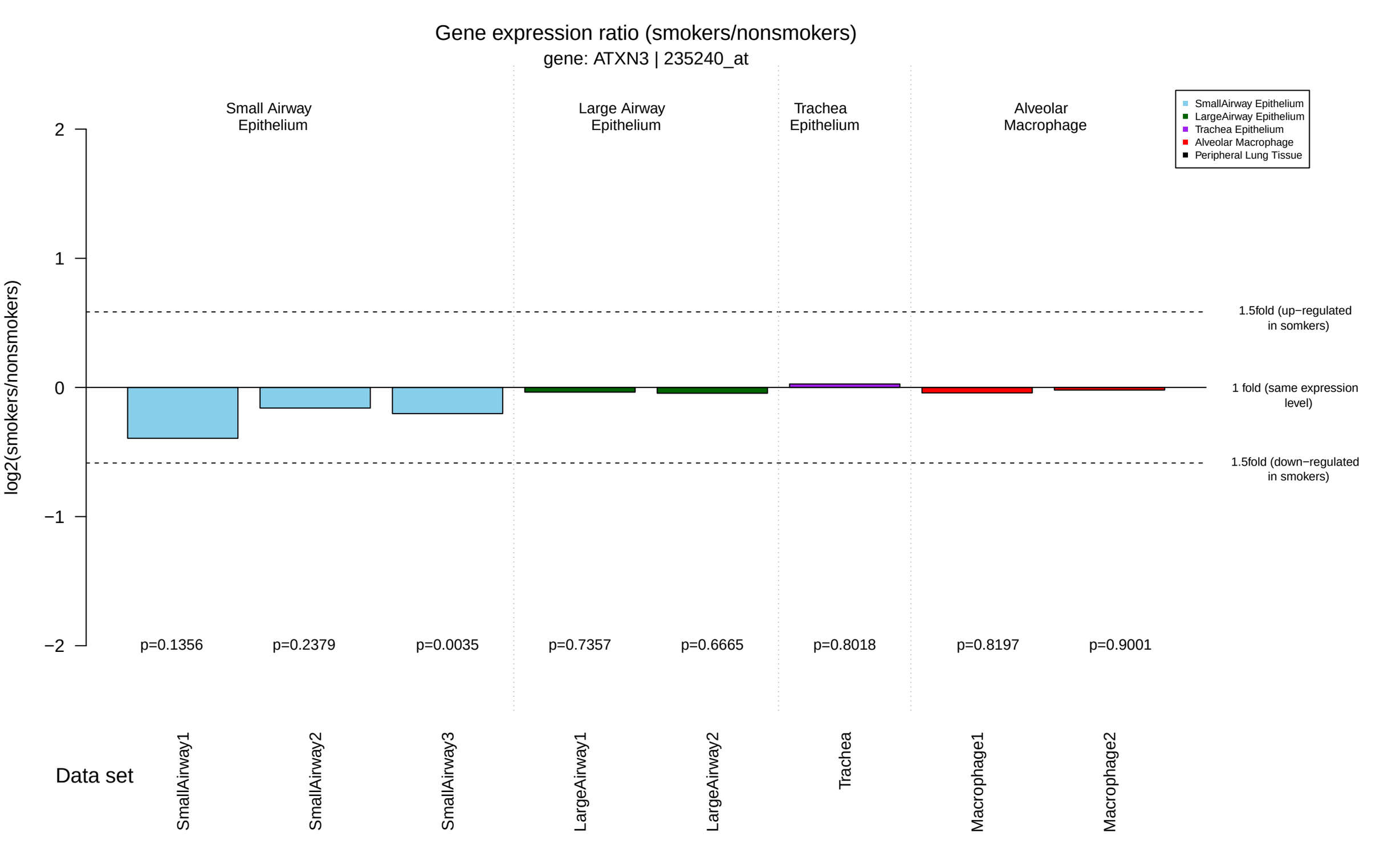
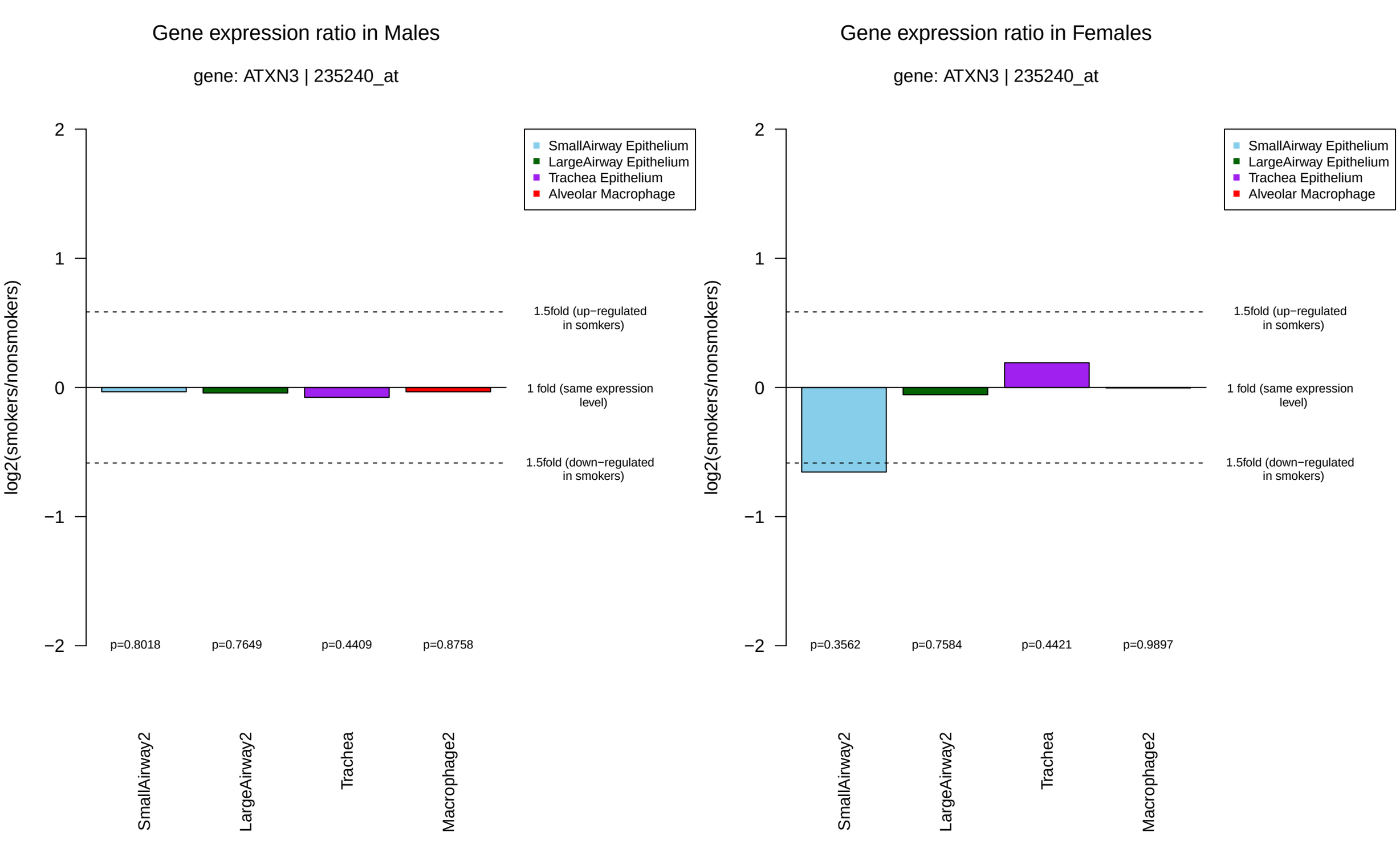
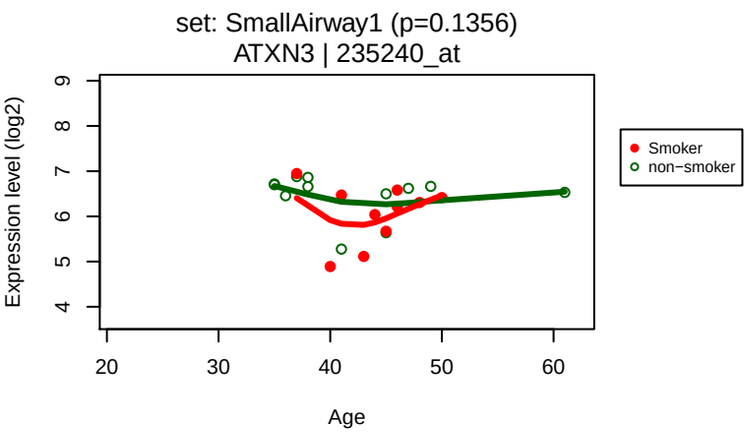
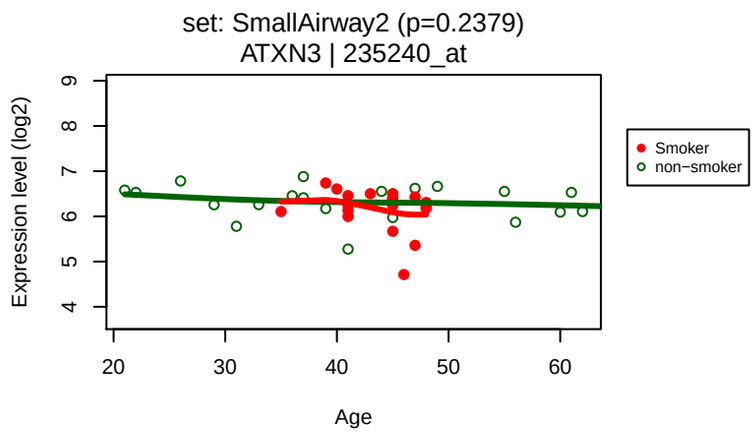
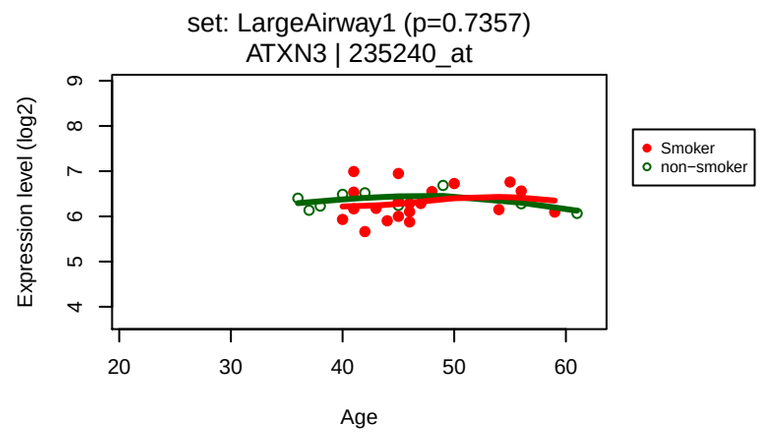

Gene expression of smokers and nonsmokers
gene: ATXN3 | 235240_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1356
p=0.2379
p=0.0035
p=0.7357
p=0.6665
p=0.8018
p=0.8197
p=0.9001
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

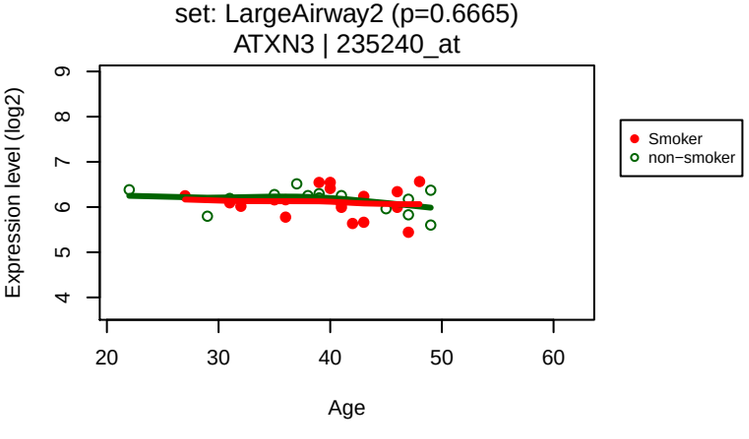

Gene expression ratio (smokers/nonsmokers)
gene: ATXN3 | 235240_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1356
p=0.2379
p=0.0035
p=0.7357
p=0.6665
p=0.8018
p=0.8197
p=0.9001
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

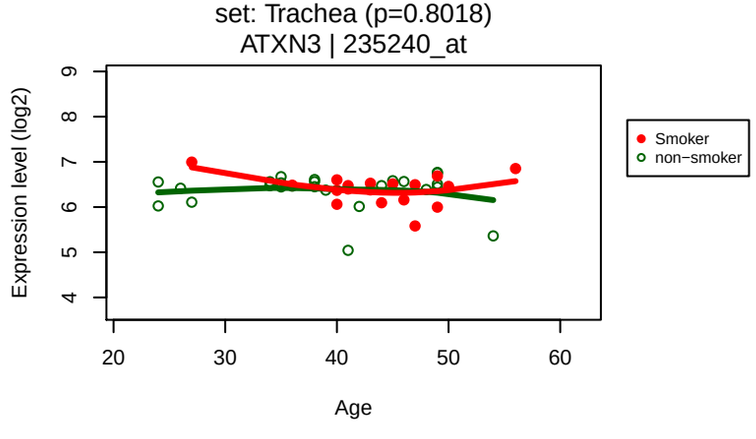

Gene expression ratio in Males
gene: ATXN3 | 235240_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8018
p=0.7649
p=0.4409
p=0.8758
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ATXN3 | 235240_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3562
p=0.7584
p=0.4421
p=0.9897
SmallAirway2
LargeAirway2
Trachea
Macrophage2

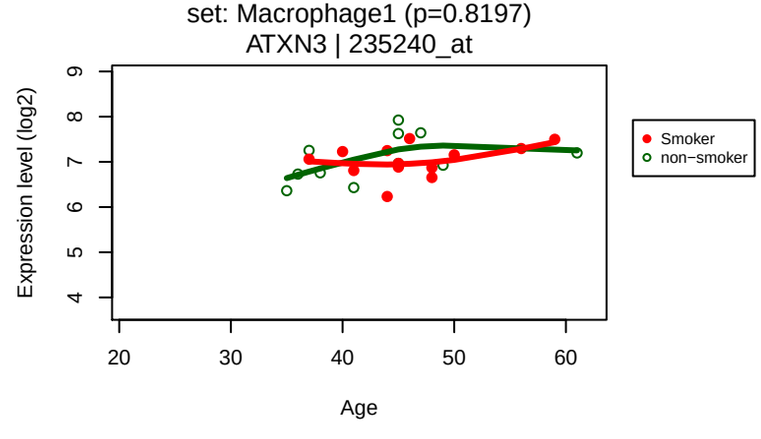

set: SmallAirway1 (p=0.1356)
ATXN3 | 235240_at
Expression level (log2)
Age
Smoker
non−smoker

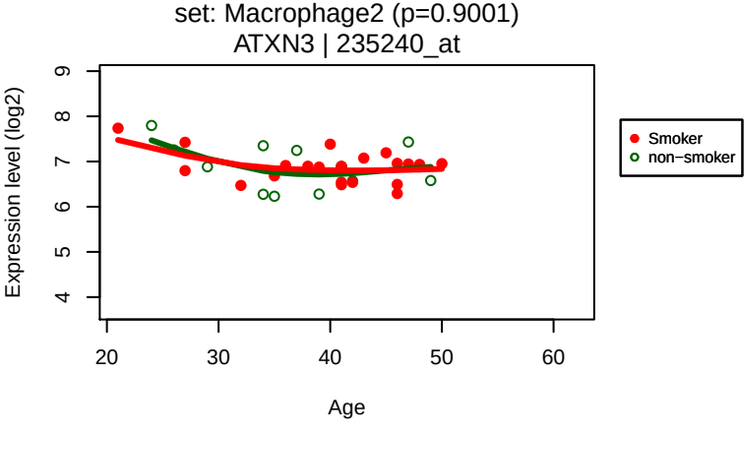

set: SmallAirway2 (p=0.2379)
ATXN3 | 235240_at
Expression level (log2)
Age
Smoker
non−smoker

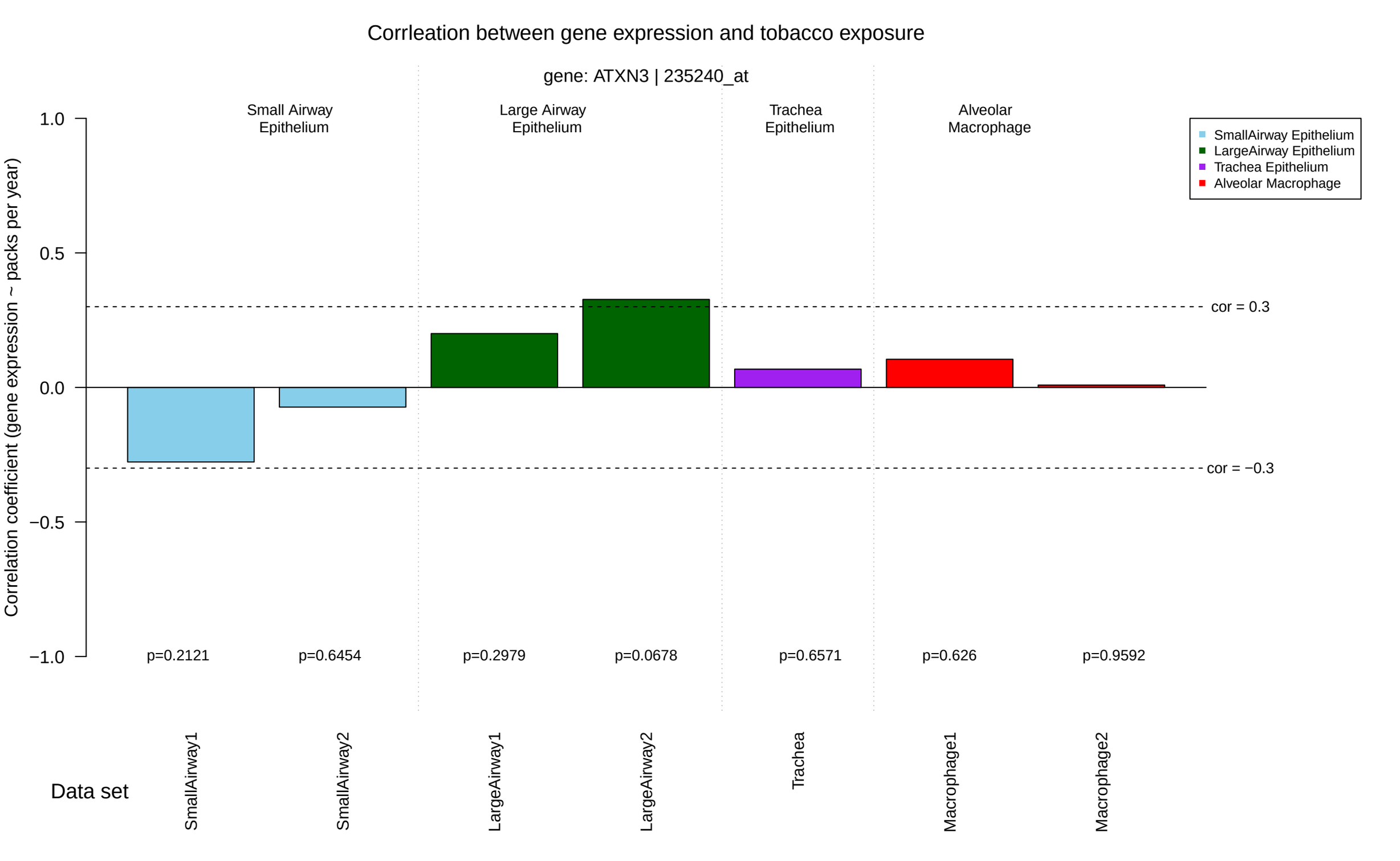

set: LargeAirway1 (p=0.7357)
ATXN3 | 235240_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.6665)
ATXN3 | 235240_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.8018)
ATXN3 | 235240_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.8197)
ATXN3 | 235240_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.9001)
ATXN3 | 235240_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ATXN3 | 235240_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.2121
p=0.6454
p=0.2979
p=0.0678
p=0.6571
p=0.626
p=0.9592
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2